



Valmet DNA Engineering Function Test Manual

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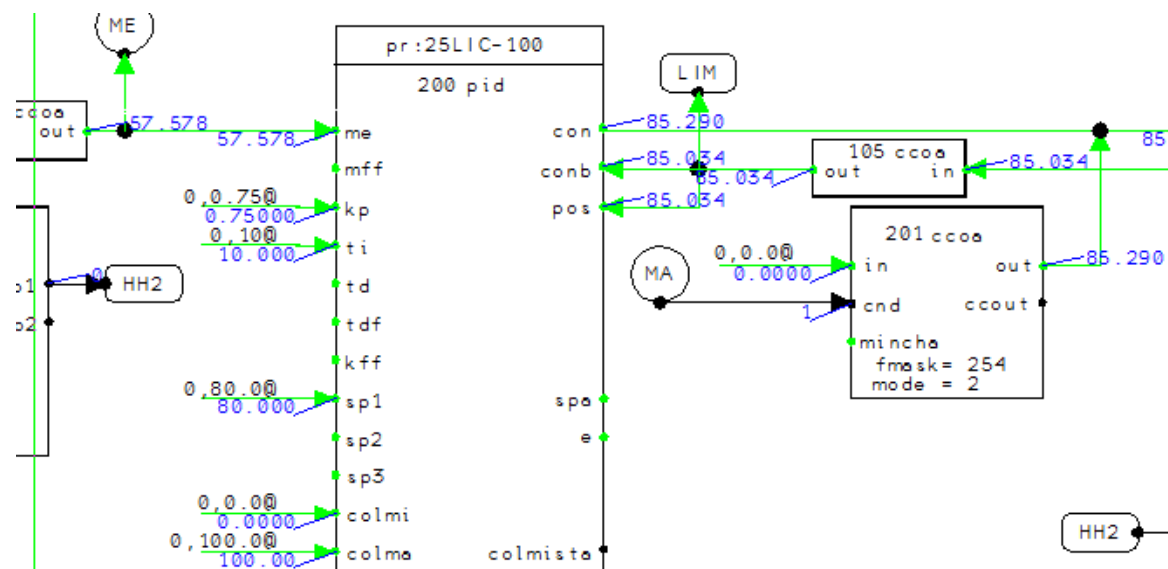
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1. Introduction

Function Test is a tool to test a function block or sequence diagram with continuously updated values from the runtime environment (real or virtual). Function Test can be used, for example, to verify the correct functionality of a diagram under design or to find a mistake if the diagram does not work the way it should.

The user opens a function block or sequence diagram in Function Test and requests the updated values to the diagram. They are shown in the connection points of the function blocks. The user can implement changes by entering new values to the connection points and the function block parameters. Additionally, the user can step the stored history values in the diagram back and forth to see how things really happened in the past.



A function block diagram showing continuously updated values in Function Test

2. How to Test a Diagram

This chapter describes a basic workflow when testing a control loop with the Function Test tool. To find out all the details and options, study thoroughly *Function Test Manual*.

In the following example, you have designed a loop in Function Block CAD and you want to check if the functions you have implemented work correctly. Proceed as follows:

1. Start the Function Test tool with the menu bar File -> Test command. Function Test is opened including the function block diagram and the Function Test toolbar.

NOTE! Function Test is now running. A separate start command is not needed.

2. Add test points by using the following toolbar buttons:

-  Insert test points one by one
-  Insert test points for given area
-  Insert all test points to current page

The test points appear to the connection points of the diagram and are updated continuously.

NOTE! You can add new test points whenever you want during the test.

NOTE! You cannot save the inserted test points in the diagram when you close Function Test. In case you have to return to the test repeatedly, it is recommended to add all test points to a current page (or to a selected area) for time saving reasons.

3. Enter new values to the test points to test the functionality of the diagram.
First click the toolbar button  and then the test point. Enter a new value in the dialog.
4. Check if the changes you made caused a correct response in other test point values.

You can also trace the changes by opening the History view. First click the toolbar History>> button. Then pause the history collection and the updating of the test values with the  button so that you can start studying the collected values. Select the desired moment in history with a slider and use the stepping buttons  to step back and forth in time. The corresponding values are updated to the diagram. You can also move the slider to a specific position by entering a cycle number to the Cycle field and then clicking the Go to button.

To add trends, see the detailed instructions in chapter 7 [Viewing Test History as Trend Curves](#).

5. Return to Function Block CAD by clicking the toolbar button Return to CAD.

NOTE! The use of Function Test does not cause any changes to the original function block diagram.

4. Changing the View

You can use the toolbar commands to change the view showing the diagram:

- Zoom area. Click the  icon and define the zoomed area by drawing a rectangle on the diagram. End the command with a click. The area selected is expanded to fill the view.
- Zoom previous. Click the  icon. The previous zoom state is shown.
- Zoom all. Click the  icon. The whole diagram with all the pages is shown.

You can also rotate the center wheel of the mouse to Zoom In and Out, and press the center wheel down to pan the diagram.

You can use menu bar command View > Select Page to select the page in the view. It also selects the page to which the commands are focused.

In the View menu you will find more commands familiar from Function Block CAD for changing the view. For more information, see chapter 12.2.3 [View - Display Handling Commands](#).

5. Viewing Updated Values as Test Points

After starting Function Test, add test points to show the continuously updated numeric values in the diagram. You can add them by using the following toolbar commands:

- Add test points to all connection points on the current page: Click .
- Add test points to the connection points inside a selected area in the diagram:
Click  and draw a rectangle on the diagram. End the command with a click. The test points are inserted inside the rectangle.
- Add test points to the selected connection points: Click  and then the desired connection points one by one. End the command with the Return key.

Inserting test points to all the connection points on the current page is a practical command because it takes only a short time to get them in the view and, therefore, you do not need to think about which test points you actually need.

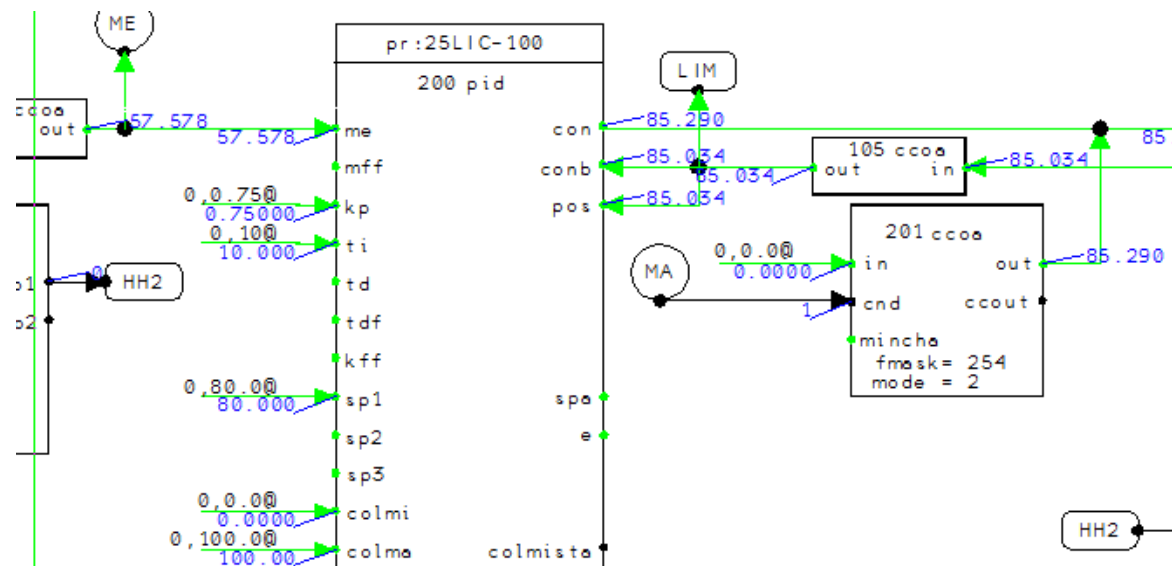
NOTE! You cannot save the inserted test points in the diagram for a later use. When you reopen the diagram in Function Test later on, there will not be any test points in the view.

From the menu bar, you will find more commands for inserting test points (Test Points > Insert Test points to). For example, you can insert test points to all the connected connection points only or to all the pages of the diagram. For more information on commands, see chapter 12.2.4 [Test Points - Test Point Managing Commands](#).

Test points can be deleted in the following ways:

- All the test points on the page with a toolbar or menu bar command: Click  icon on the toolbar or select the Test Points -> Clear All Test Points menu bar command.
- The selected test points with the Delete key: Select the test points and then press the Delete key.

Test points are shown as blue numeric values with the handle connected to the connection point of the corresponding function block (see the next figure).



Test points of a function block

Values are updated to the test symbols from the runtime environment. The active fault bits related to these values will be expressed by displaying the sum of fault bits next to the actual value. For example:

3.1415+(136)

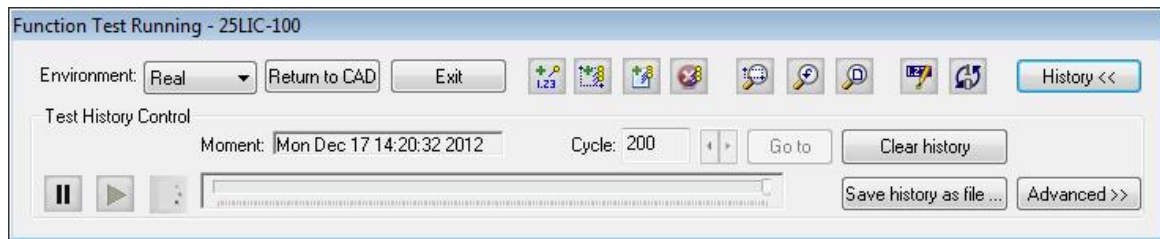
The above example contains the analog value 3.1415 where the fault bits *Dis* and *Sex* are set on (128+8=136). Fault bits you can find out from the dialog which is

opened with the toolbar  button or the Test Values -> Enter New Value menu bar command. For more information on the dialog, see chapter 8.1 [Entering Test Point Values](#).

6. Viewing Test History as Test Point Values

You can expand the toolbar with the toolbar button History >>. The expanded area contains controls to manage the test history.

Function Test collects the history from all the test points added to the diagram. The test history is shown as numeric values in the test points. With the history controls, you can trace a phenomenon which is hard to perceive at a glance. You can also save values in a file for performing a more detailed analysis.



Test history controls in the Play state

Play state (History collection)

When Function Test is opened, it is in the *Play* state by default. This means that it starts collecting history from the test points as soon as they are inserted to the diagram. The test values are continuously updated to the diagram.

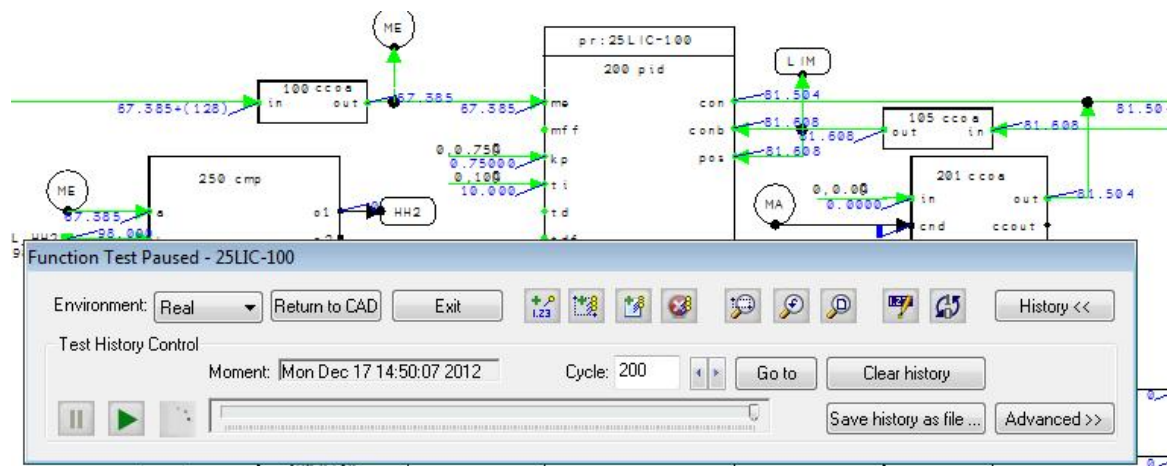
When Function Test is collecting history, the Play state indicator  is rotating. The slider, the Moment field and the Cycle field indicate the time and the collection history location of the current value.

You can clear the history and reset the collection to start from the first cycle by using the Clear history button or the menu bar command Test Control -> Clear Test History.

You can define the collection interval in milliseconds (Test interval) and the length of the history in cycles (Maximum length of history) in the advanced settings opened with the Advanced>> button.

Pause state (Analyze)

You can pause the history collection and the updating of the test values with the  button and start studying the collected values. You can move the slider to a specific moment (or cycle) and step it back and forth with stepping buttons . You can also enter a cycle number to the Cycle field and click the Go to button. The test points in the diagram will show the corresponding values from the selected moment in history.



Test history controls in the Pause state, selected moment and the corresponding test values from the same moment

Click the Play button  to restart the history collection. Function Test starts updating the test values in the diagram and collecting history starting from the current values.

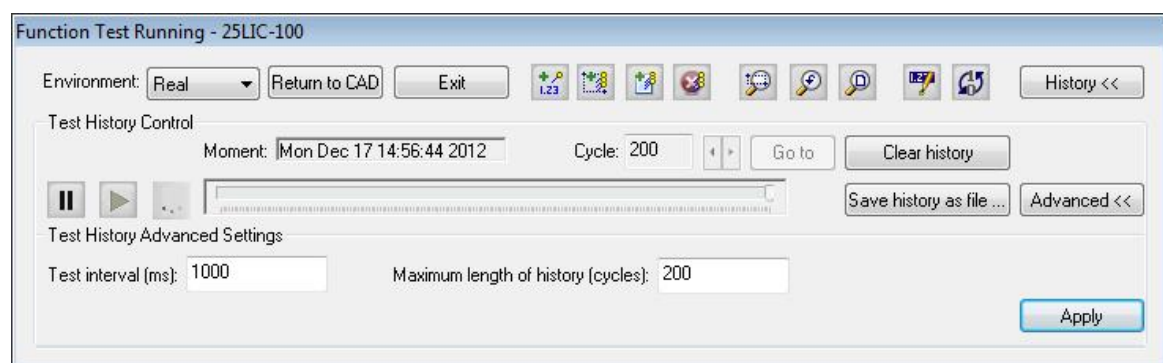
Note that you can find Play and Pause commands also from the Test Control menu in the menu bar.

Save history

You can save the history as a text file with the toolbar Save history as file button or with the menu bar command Test Control -> Save Test History as File in order to view all the numeric values from all the variables at one go, or to move the values to Excel a for further analysis, for example.

Advanced history settings

You can open the Test History Advanced Settings section with the Advanced >> button and set the updating interval and length of the history for history collection.



Advanced test history controls

Note that the updating interval defines both the collection interval of the history and the updating interval of the display. When the diagram is opened in Function Test, the interval is automatically set to match the execution cycle of the function block diagram.

You can open the Test History Advanced Settings section with the Advanced << button.

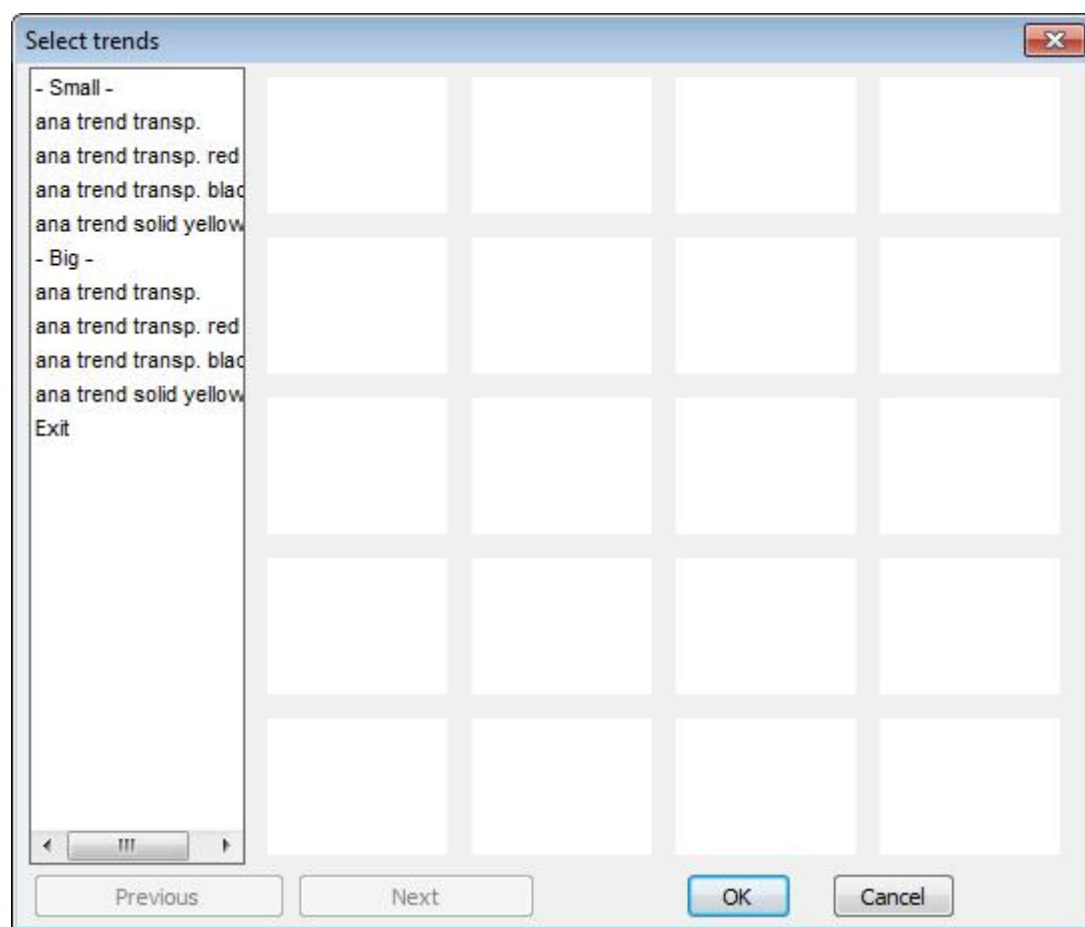
7. Viewing Test History as Trend Curves

You can view the history of the selected values as trend curves. Trend curves are useful to show interdependencies and timing in a graphic way.

For each trend curve, you create a background with a scale and then you insert a curve in it. However, by using transparent backgrounds you can place several trend curves on top of each other.

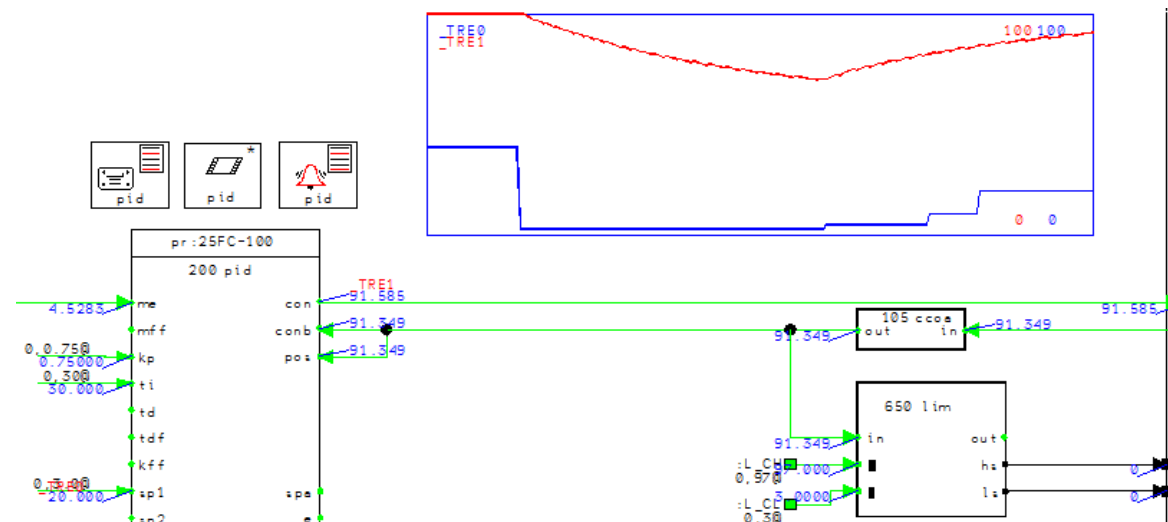
7.1 Inserting a Trend Area

Select the Trends -> Create Trend command from the menu bar. A dialog where you can select the appearance of a trend is opened.



A dialog where you select the appearance of a trend

In the dialog, you can select the background of the trend (transparent or solid yellow), the trend curve colors (blue, black or red trend curve) and the size of the trend (small or big).



An example of two big transparent trend curves on top of each other

After clicking the OK button, place the trend area at the location you want. A dialog for inserting the scale is opened asking the minimum and maximum values for the curve.

NOTE! You do not need to enter a variable name in the dialog because the name is automatically added when you insert a signal for the trend curve later on. However, if you have earlier designated a name (with the Edit -> Values command) for the test point which will be inserted to the trend area, and you want to use the same name in a trend, you can enter the name in the Variable name field of the dialog.

7.2 Inserting a Test Point to a Trend Area

Next, you have to insert a test point to the created trend. Select the Trends -> Insert Signal as Trend Curve command from the menu bar. First click the test point in the diagram and then the trend area. The trend starts to update if the Function Test is in the *Play* state. The name of the trend is shown in the trend area and beside the corresponding test point.

If a table-type signal has been selected to a trend area, a profile consisting of signal values is displayed instead of a trend.

7.3 Making Changes

You can change the scale of the trend with the menu bar command Edit -> Values. You can move and resize the trend with the Move and Resize Trend commands in the Trends menu. However, when using these commands, Function Test must be in *Pause* state.

You can delete an individual trend with the Delete key, and all the trends with the menu bar command Trends -> Remove All Trends.

Tips:

If you have placed several trends on top of each other, you can select the specific trend by pointing at the name or scale of the trend.

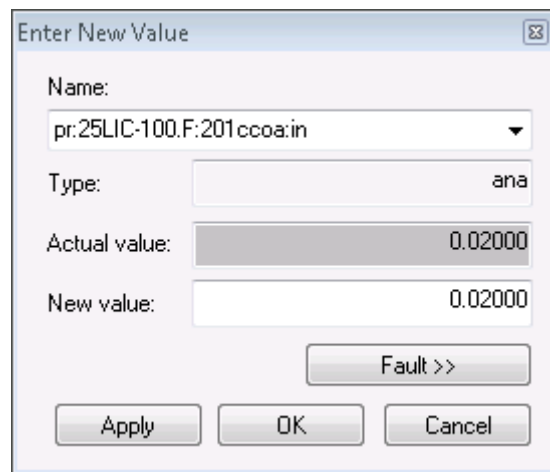
When you add a new trend area and a curve on top of another, they are not synchronized: the current values are shown in different locations in the trend area. To get them synchronized, reset the history or wait until the current values reach the right edge of the trend area.

8. Entering a New Value

To test the functionality of the diagram, you can change the values in the runtime environment. With the toolbar command  or the menu bar command Test Values -> Enter New Value, you can change the values of analog, binary and integer type test points and function block configuration parameters. Depending on the type of the value, different kinds of dialogs open for entering the value.

8.1 Entering Test Point Values

Click the toolbar command  or the menu bar command Test Values -> Enter New Value, and then the desired test point. A dialog for entering a new value is opened. Enter the new value for the test point in the New value field in the dialog and click the Apply or OK button.



Enter New Value dialog for an analog value

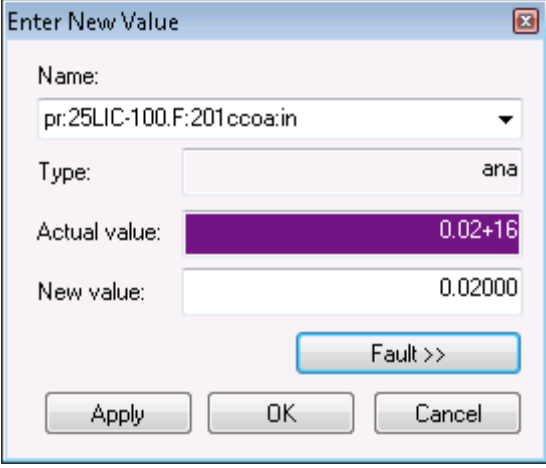
You can change the name of the target test point value by:

- typing a new name into the Name field. The field can also address outside the viewed diagram, for example to another function block diagram being tested.
- selecting a name from the Name drop-down list box showing the names recently used.

Tip:

There can be several dialogs open on the screen at the same time so that you can enter new values for various test points and function block configuration parameters.

Status information of the value is shown by the background of the current value. If a fault bit *Sex* or *Dis* is active (set on), the background is colored with the warning color (yellow). Other faults are indicated with the fault color (purple).

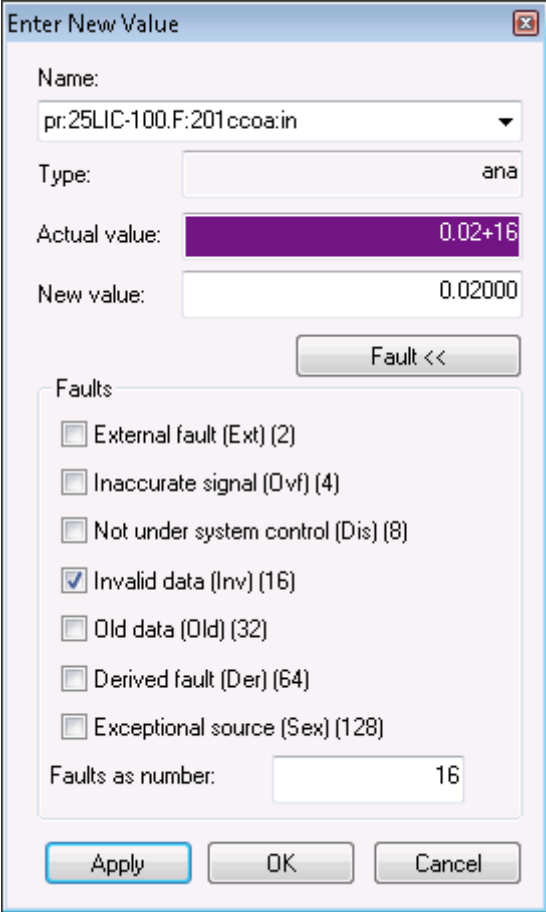


The dialog box is titled "Enter New Value". It contains the following fields and buttons:

- Name:** A dropdown menu showing "pr:25LIC-100.F:201cco:in".
- Type:** A text field containing "ana".
- Actual value:** A purple bar with the text "0.02+16".
- New value:** A text field containing "0.02000".
- Buttons:** "Apply", "OK", "Cancel", and a highlighted "Fault >>" button.

Enter New Value dialog with an active fault

More detailed information about faults can be retrieved by clicking the Faults>> button. It opens the Faults section through which the user can see the cause of the fault and also set a new fault status by checking its check box.



The dialog box is titled "Enter New Value". It contains the following fields and buttons:

- Name:** A dropdown menu showing "pr:25LIC-100.F:201cco:in".
- Type:** A text field containing "ana".
- Actual value:** A purple bar with the text "0.02+16".
- New value:** A text field containing "0.02000".
- Buttons:** "Apply", "OK", "Cancel", and a "Fault <<" button.
- Faults section:** A list of fault types with checkboxes:
 - ☐ External fault (Ext) (2)
 - ☐ Inaccurate signal (Ovf) (4)
 - ☐ Not under system control (Dis) (8)
 - ☒ Invalid data (Inv) (16)
 - ☐ Old data (Old) (32)
 - ☐ Derived fault (Der) (64)
 - ☐ Exceptional source (Sex) (128)
- Faults as number:** A text field containing "16".

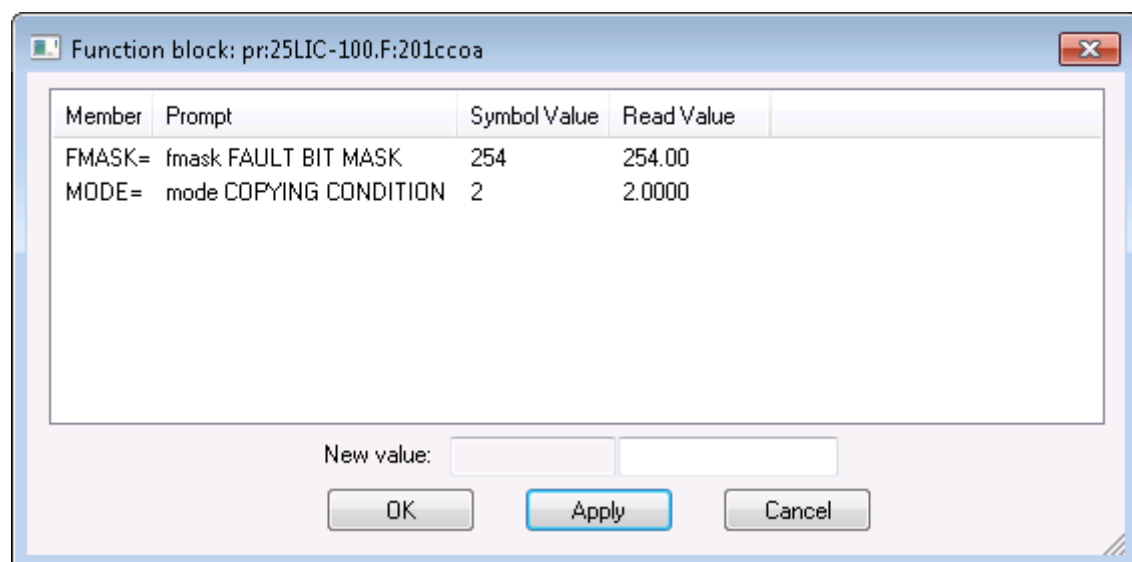
Enter New Value dialog with Faults section opened

You can invert the value of a binary state with the toolbar command . Click the  button and select the binary-type test point. The value is changed from 0 to 1 or from 1 to 0. You can also invert by using the menu bar command Test Values -> Invert Binary Values.

If you know the name of the test point the value of which you want to modify (for example a test point of another control loop), you can open the same Enter New Value dialog with the menu bar command Test Values -> Enter Value by Name command and enter the new value through it.

8.2 Entering Function Block Configuration Parameter Values

Click the toolbar command  or the menu bar command Test Values -> Enter New Value, and then click a function block symbol. A dialog for entering a new value is opened. Select the configuration parameter to be changed by clicking its name. Enter a new value for the configuration parameter in the New value field in the dialog and click the Apply or OK button.



Enter New Value dialog for a function block configuration parameters

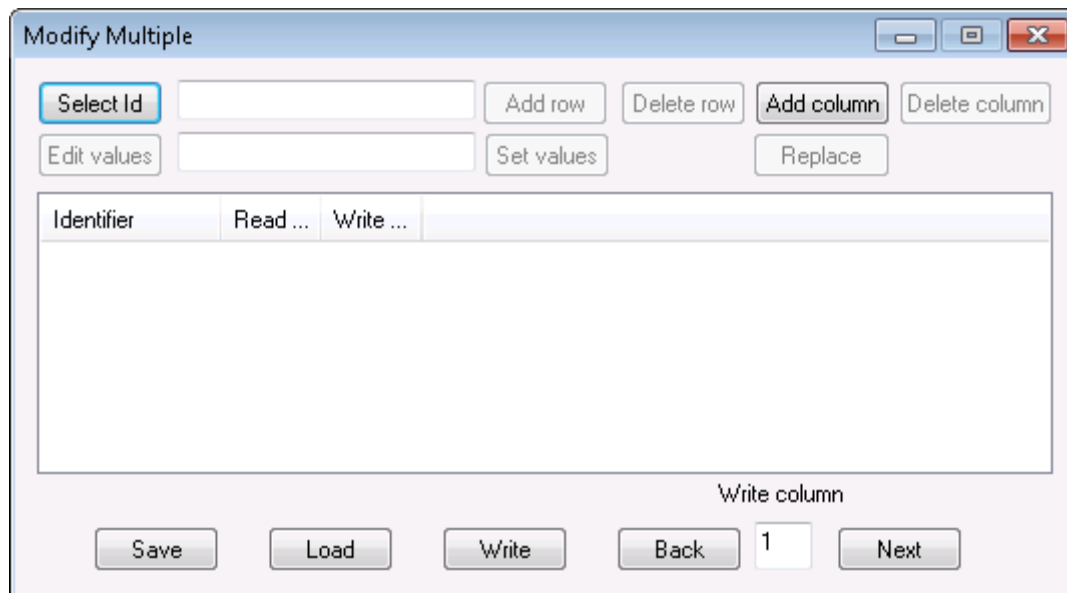
Tip:

There can be several dialogs open on the screen at the same time so that you can enter new values for various test points and function block configuration parameters.

9. Entering Multiple Values Simultaneously

You can enter multiple values at the same time. This is a useful feature when the timing of inputs is critical, or when you want to perform a test sequence with predefined sets of values.

Select the menu bar command Test Values > Enter Multiple Values. The dialog shown below is opened.



Modify multiple dialog

9.1 Filling in a Table of Values

In the Modify Multiple dialog, you can fill in a table where rows represent different test points and the Write value columns represent different sets of values to be downloaded into the runtime environment. After you have finished filling in the table, you can write all the values in each column simultaneously to the runtime environment by clicking the Write button.

First, select a test point from the diagram and then click the Select Id button. The name of the test point appears in the Select Id field. Then, click the Add row button. The test point is now shown as a new row in the table. Repeat this until all the test points to be modified are in the table. Only the test point name (Identifier) and the current value (Read value) are shown on each row.

The table also contains a column for new values. Select the test point name in the table by clicking it, enter a new value into the Edit values field and click the Set values button. A new value appears in the Write value column of the table.

Modify Multiple

Select Id: pr:25LIC-100.F:200pid:sp3

Edit values: 77

Buttons: Add row, Delete row, Add column, Delete column, Set values, Replace, Save, Load, Write, Back, Next

Identifier	Read value	Write value 1
pr:25LIC-100:sp1	80.000	75
pr:25LIC-100.F:200pid:sp2	68.000	76

Write column: 1

Modify Multiple dialog, entering multiple values simultaneously

9.2 Writing the Values to the Runtime Environment

When the Write value column is ready, click the Write button in order to enter the values. All the values in Write value column are sent to the runtime environment and the new values are updated to the Read value column.

9.3 Working with Multiple Sets of Test Values

Use the Add column button to add more columns to be filled in. The Write column field in the bottom right corner always shows to which column the Write command is directed. You can change the active column with the Back and Next buttons.

To enter new values for multiple columns, select the modified row by clicking the test point name (Identifier). Then, enter new values for all the columns into the Edit values field. The values are separated by commas, for example 76,77,99. Click the Set values button to move the values into the table.

NOTE! You are allowed to select another row and store the same values to its columns too by clicking the Set values button again.

Identifier	Read value	Write value 1	Write value 2	Write value 3
pr:25LIC-100:sp1	80.000	76	77	99
pr:25LIC-100.F:200pid:sp2	75.000	76	77	99

Modify multiple dialog, entering multiple sets of values simultaneously

When you edit the values in the table later on, you can use the Edit values button to copy the values from the selected row into the Edit values text field.

You can use the Delete column button to remove the last column added and the Delete row button to delete the selected row.

9.4 Saving the Table of Values for a Later Use

You can save the contents of the table to a file with the Save button and read them back from a file to the table with the Load command.

If you want to use the same saved test point values, for example, in a different diagram with different tag names, you can use the Replace button to modify test point names. First, use the Load button to read the values from the file to the table. Then open the Replace dialog with the Replace button, enter the value to be retrieved, and the new, substitutive value in the dialog. Finally click the Replace button.

Replace dialog

NOTE! Replace command only modifies test point names.

10. Controlling the Test with Automatic Start and Stop

You can control the testing and the history collection by using automatic start and automatic stop commands in the Test Control menu. They add the corresponding functions into the diagram. They allow you to define the triggering test signals and related conditions to start or stop the history collection (e.g. $A > 1$). Automatic start and Automatic stop functions are useful when you trace a fault that occurs very seldom. You can leave the test running, and when the defined condition is true, Function Test freezes the situation to be studied later on.

NOTE! When you add the Automatic start function, the history collection is stopped immediately if the start condition is not yet true (usually it is not). However, the test values are updated in the diagram. Automatic stop pauses both the history collection and the updating of the test values in the diagram.

You can insert the Automatic start into the diagram through the Function Test menu bar command Test Control -> Set Automatic Start. A dialog, like the one shown below, opens.

Prompt	Value
Start condition	A==1
Delay (in steps)	0
*Value of function	1
*Start status	

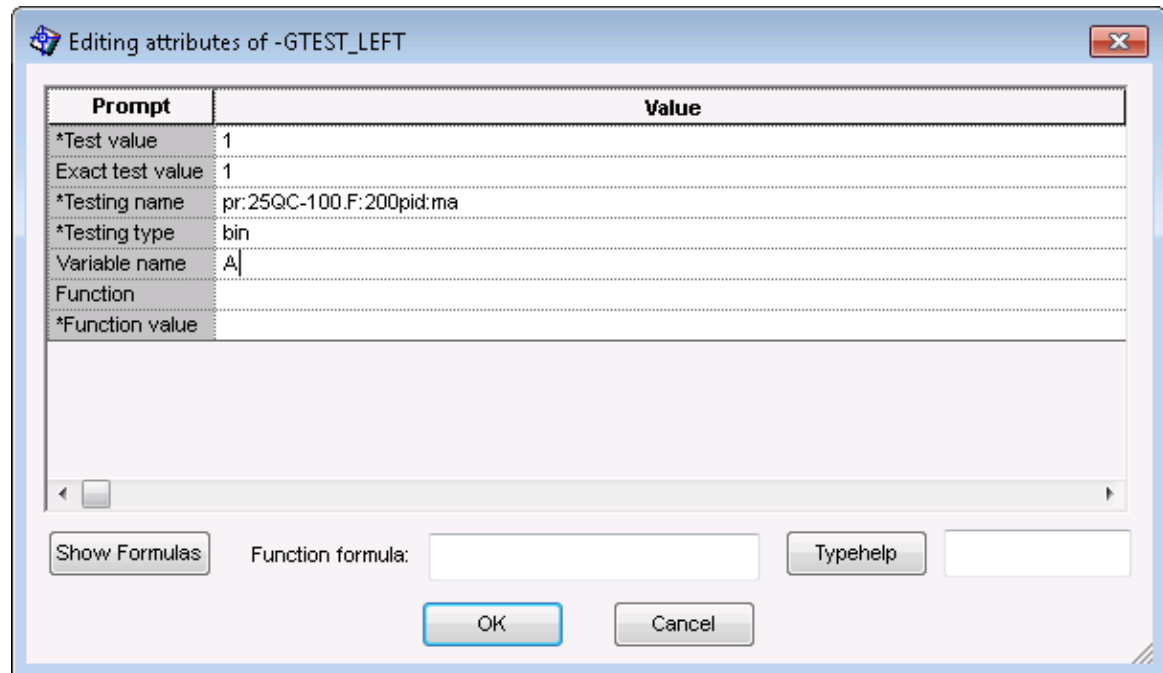
Function formula:

Buttons: Show Formulas, Typehelp, OK, Cancel

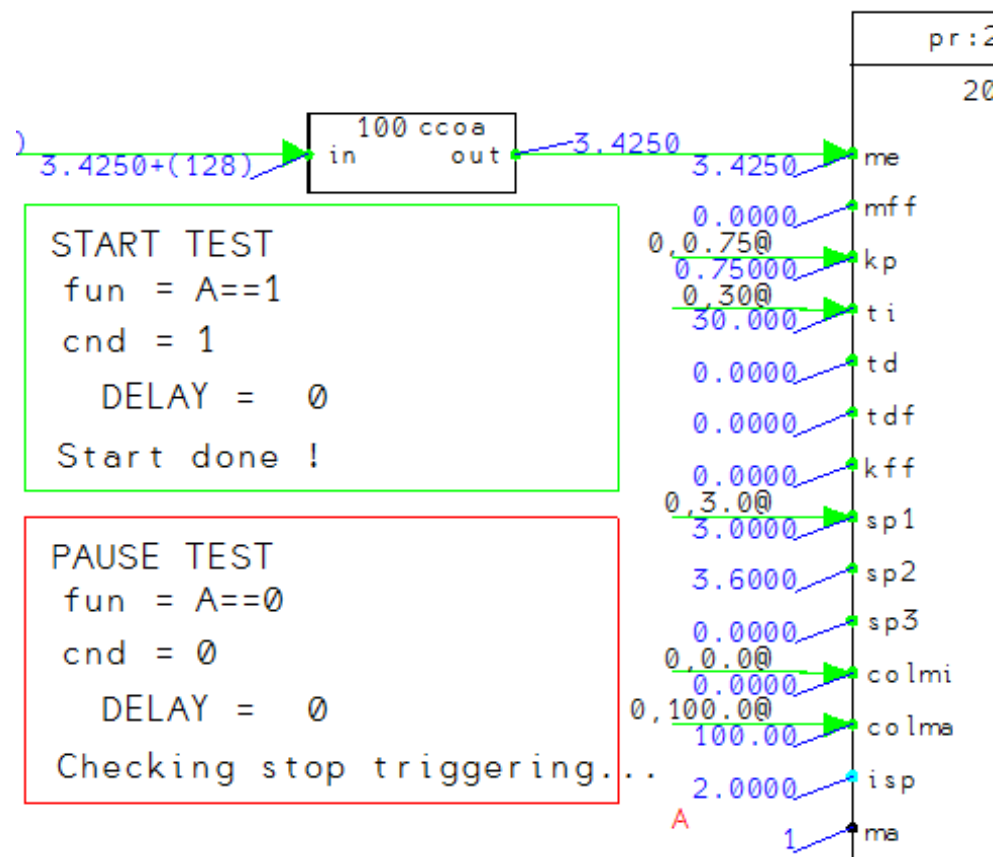
Automatic start symbol dialog

In the dialog, you enter the attributes Start condition (condition for triggering) and Delay (after triggering). The delay is given in test steps. Allowed operators in the conditions are: $<$, $>$, $<=$, $>=$, $!=$ and $==$. In addition, the Value of function attribute shows whether the condition is being met at the moment (*Value of function=1*) or not (*Value of function=0*) and the Start status attribute shows the test status in text format (for example, *Stop done!*). The same status information is also shown in the *Automatic start* symbol.

The variable name used in the condition must be defined in the test symbol dialog that opens through the Edit->Values menu bar command. The Automatic stop dialog is filled similarly as the Automatic start dialog.



The *Automatic start* and *Automatic stop* symbols are shown in the testing tool as in the figure shown below:



An example of automatic start and automatic stop symbols

In the example figure, the test is started instantly ($DELAY=0$) when the controller's ma value is 1 and stopped without delay when the ma value is 0. The variable name A has been defined for the test symbol of the controller's ma value.

NOTE! You can add several automatic start and stop functions in one diagram.

11. Exit Function Test

You can exit the Function Test by

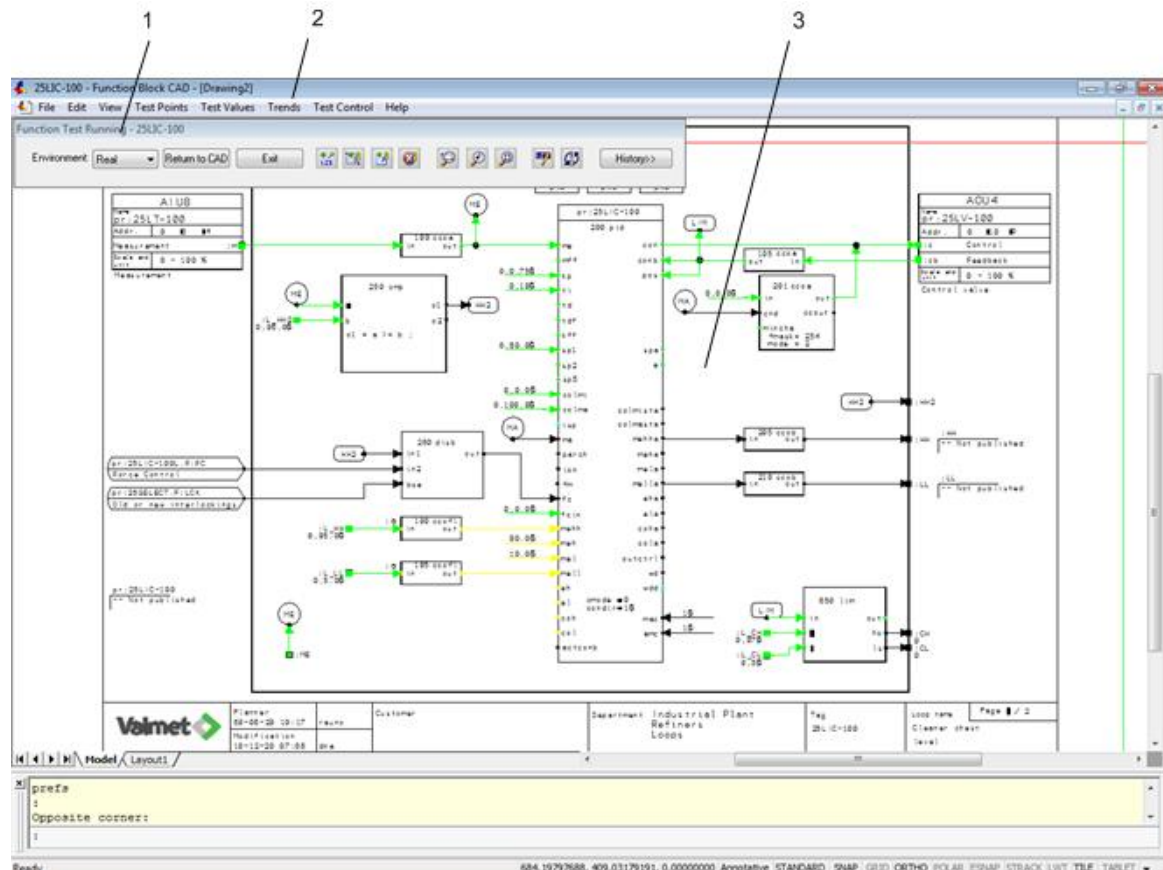
- returning to Function Block CAD or Sequence CAD with the toolbar button Return to CAD or with the menu bar command File -> Return to CAD.
- closing CAD tools completely with the toolbar button Exit or with the menu bar command File -> Exit Function Tool.

No test points or any history is saved in the function block or sequence diagram when Function Test is closed.

12. Functions of User Interface

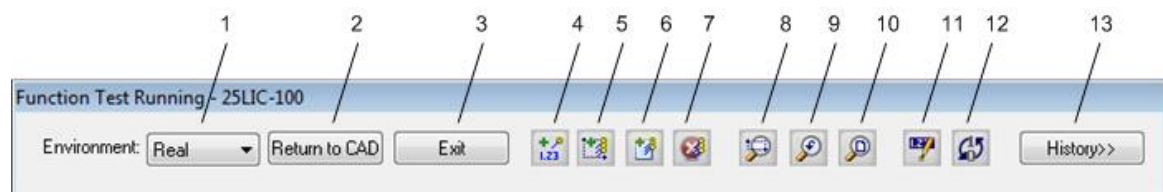
Function Test user interface consists of the following parts:

- Function block diagram or sequence diagram (3)
- Function Test toolbar (1)
- Function Test menu bar (2)



Function Test user interface parts

12.1 Toolbar Commands



Function Test toolbar

The controls of the toolbar are:

1. The current runtime environment
2. Exit Function Test and return to Function Block or Sequence CAD.
3. Exit Function Test and close Function Block or Sequence CAD.
4. Insert test points to the selected connection points.
5. Insert test points to the indicated area.

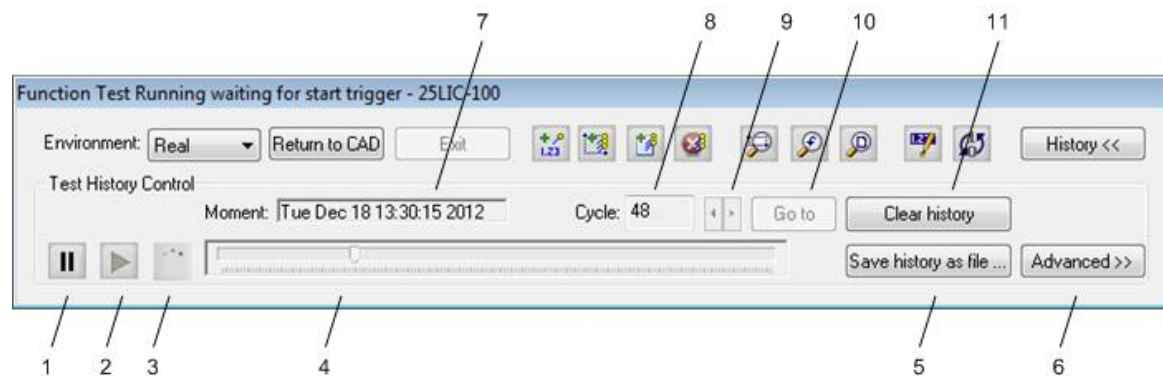
6. Insert all test points to the current page.
7. Remove all test points from the diagram.
8. Zoom in the selected area.
9. Restore the previous view.
10. Show the whole diagram including all pages.
11. Enter new value to a selected test point.
12. Invert the state of a selected binary value.
13. Open the Test History Control section.

Guidance is given as a tooltip when you move the cursor over a specific control in the toolbar.

In the Environment field of the toolbar, you can select either the real or virtual runtime environment to be the testing environment.

12.1.1 Test History Control Section of the Toolbar

With the toolbar History >> button, you can open the Test History Control section the functions of which you can use to manage the test history.



Test History Control section of the toolbar

The controls of the Test History Control section are:

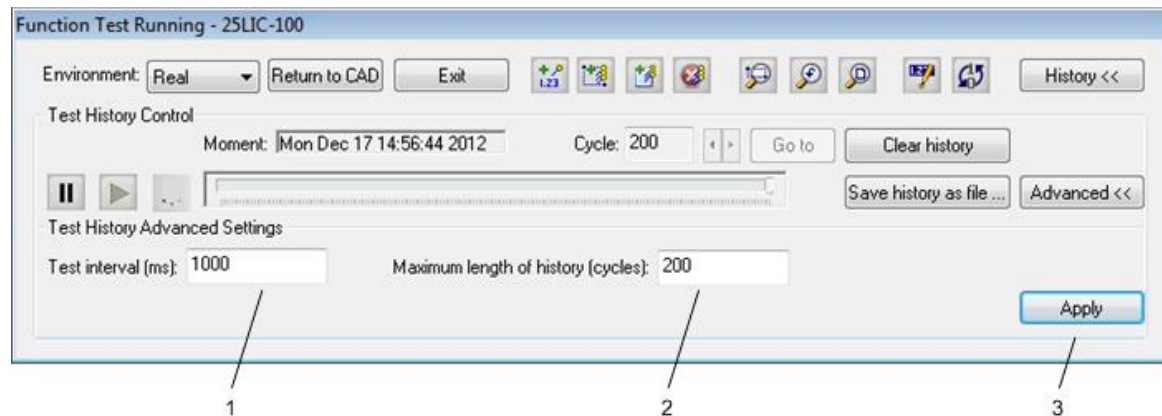
1. Pause the history collection and the updating of the test values.
2. Restart the history collection starting from the current values.
3. Play state indicator (rotates when history is collected).
4. Slider (which indicates and with which you can select the moment of history).
5. Save the history as a text file.
6. Open the Test History Advanced Settings section.
7. Time of the displayed history values (the moment indicated by the slider)
8. Position in collection history for the displayed values
9. Step back and forth the specific moment or cycle indicated by the slider (available only in *Pause* state).
10. Move to the moment in history which the Cycle field indicates (available only in *Pause* state).
11. Clear the history and reset the collection to start from the first cycle.

Guidance is given as a tooltip when you move the cursor over a specific control.

For more information on the functions, see chapter 6 [Viewing Test History as Test Point Values](#).

12.1.2 Test History Advanced Settings Section of the Toolbar

With the Advanced >> button of the Test History Control section, you can open the Test History Advanced Settings section.



Test History Advanced Settings section of the toolbar

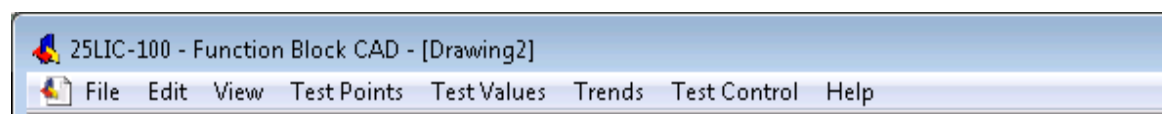
The controls of the Test History Advanced Settings section are:

1. Define the updating interval of the history collection and display.
2. Define the length of the history for history collection.
3. Use the changed information (updating interval and length of history collection) in testing.

Guidance is given as a tooltip when you move the cursor over a specific control.

For more information on the functions, see chapter 6 [Viewing Test History as Test Point Values](#).

12.2 Menu Bar Commands



Function Test menu bar

Some of the commands in the menu bar are also available in the toolbar.

12.2.1 File - File Managing Commands

Print (Pause required)...
Print Setup (Pause required)
Navigate
Return to CAD
Exit Function Test

File pull-down menu

Print (Pause required)...

Prints a diagram with all the test information to a chosen printer. Select one of the printing functions (*display, all pages, page or range*). Note that you can use this command only when Function Test is in *Pause* state.

Print Setup (Pause required)

Settings for the printer: select the printer. Note that you can use this command only when Function Test is in *Pause* state.

Navigate

The Navigate command can be used to clarify the connection chains between an external data point and various ports. It can also be used to open any design function displayed in the Navigate dialog for testing, for example a composite function block. Note that the Navigate command will only function in function block and sequence diagrams which have been identifier checked and saved in the repository. The Navigate command can be used in navigating through direct access or interface port, function block port or external datapoint.

Navigate command is also needed when you want to test a composite function block. In which case you must first open the diagram which is using the composite function block to the Function Test. Then select the menu bar command File -> Navigate and the composite function block to be tested. The selected composite function block will be opened to a new Function Test window to be tested.

The Navigate command works like the one in Function Block CAD. See more information in *Function Block CAD Manual*.

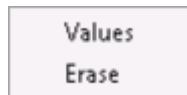
Return to CAD

Exits the testing tool and returns to the engineering tool.

Exit Function Test

Exits from both the testing and the engineering tools.

12.2.2 Edit - Editing Commands



Edit pull-down menu

Commands in the Edit pull-down menu work like those in Function Block CAD (see *Function Block CAD Manual*).

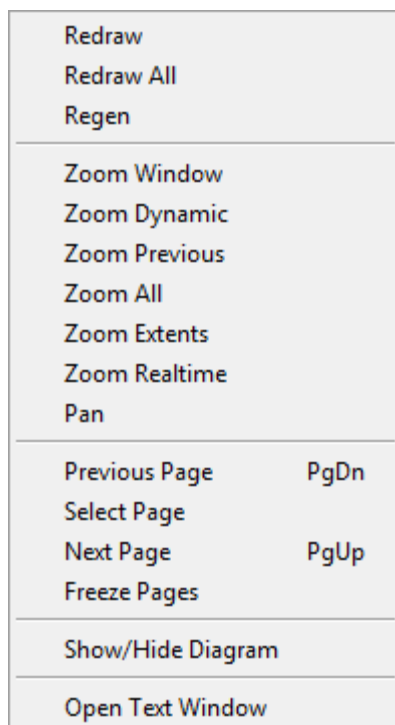
Values

By selecting this command, you can inspect and modify both the diagram symbols and the test points. The editing of attributes is done in a dialog window.

Erase

Deletes the selected test points.

12.2.3 View - Display Handling Commands



View pull-down menu

The View pull-down menu contains commands related to handling the display.

Redraw

Redraw command redraws the displayed diagram again. The command is useful for cleaning up any "remains" left by CAD from the display.

Redraw All

Redraw command affects to the window being used. If all windows need to be redrawn use Redraw All command.

Regen

Regen command clears the display and redraws the displayed diagram again. This command is useful if you have been doing a lot of editing, and there is too much "remains" in the diagram.

Zoom Window

The Zoom Window command lets you pick an area to be magnified. After selecting this command you will see the following question on the command line:

First corner:

Point the mouse at the desired spot in the diagram, and select the first corner with the left mouse button. After that you will be asked for the other corner:

Other corner:

Move the mouse cursor to another point and select this opposite corner with the left mouse button. The display will now be cleared and the area zoomed over the whole screen.

If a wheel mouse is used, the function block diagram can be zoomed also by rolling the wheel.

Zoom Dynamic

Zoom Dynamic command gives you a "snapshot" of the whole drawing area and lets you select an area of desired size for a closer examination.

To adjust the size of the area click the left button and start moving the mouse. Click the left mouse button again to accept the size and press the right button to zoom in the area inside the white frame.

Red corners on the display show the borders of the drawing. If you go beyond these borders, the diagram will be generated anew.

Zoom Previous

Zoom Previous command allows you to view the previous view. Valmet DNA always memorizes a few views, so by repeating this command you can, for instance, revoke the view before the three last zooming operations. Do not use this command for page selecting.

Zoom All

The Zoom All command brings the whole diagram to the display. If it is a multi-page diagram, you will see all pages in the same picture.

Zoom Extents

The Zoom Extents command brings all objects in the diagram to the display as large as possible.

Zoom Realtime

The Zoom Realtime command enables real time zooming. By keeping the left mouse button down and moving the mouse in up/down direction, the user can zoom the diagram in real time. Real time zoom mode is ended by pressing the Enter or Esc key.

Pan

With the Pan command you can scroll the diagram.

To scroll the diagram, select Pan in the menu bar with the left mouse button. The arrow cursor changes to a hand cursor to show that the Pan command is selected. Click the left mouse button to show the pan base point. Then move the mouse to the desired direction and click the left mouse button to move the diagram to the same direction. If you want to repeat the pan function, click the right mouse button.

If a wheel mouse is used, the diagram can be scrolled also by keeping the wheel down and moving the mouse.

Previous Page

With the Previous Page command you can move to the previous page i.e., activate it. The same function can be done with the keyboard's Page Down key.

Select Page

The Select Page command lets you move to the desired page and activate it. You will be asked for the page:

Page number? (1- n)

(where n is the total number of pages in the document).

Next Page

With the Next Page command you can move to the next page i.e., activate it. The same function can be done with the keyboard's Page Up key.

Freeze Pages

The Freeze command lets you "freeze" all pages in the diagram except the one that is currently displayed. The tool will not regenerate the frozen pages, for example, when zooming the diagram, which will speed up commands requiring a regeneration.

When you activate a frozen page using the Select Page command, the page will "melt" automatically while other pages remain as they were.

Show / Hide Diagram

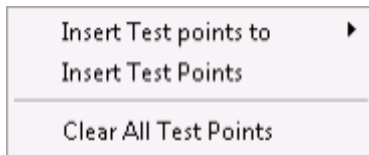
By the Show / Hide Diagram command, you can toggle the visibility of the *function*-type design layers; to show them, or not to show them. The test points are kept visible, and the visibility of the function block or sequence diagram may be changed. By hiding the diagram, a clarified view into the test points can be achieved.

Open Text window

This command brings the text window to the display. Text window is an extended version of the command line. The text window shows messages and you can give commands as well as edit them. It is possible to copy commands from the command history of the text window and paste them on the command line to be executed. These Copy and Paste commands can be selected by clicking the right mouse button on the command line.

The text window can be brought to the display also with the F2 key.

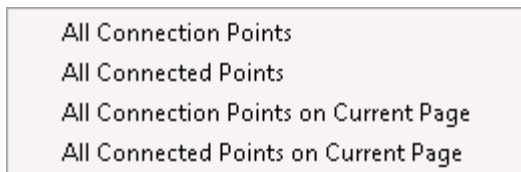
12.2.4 Test Points - Test Point Managing Commands



Test Points pull-down menu

Insert Test points to

The Insert Test Points to command opens a submenu with commands for inserting test points:



Insert Test Points to submenu

All Connection Points

Adds test points to all connection points in all pages of the diagram.

All Connected Points

Adds test points to all connected connection points in all pages of the diagram.

All Connection Points on Current Page

Adds test points to all connection points in the displayed page of the diagram.

All Connected Points on Current Page

Adds test points to all connected connection points in the displayed page of the diagram.

Insert Test Points

The Insert Test Points command inserts test points to the selected connection points. After selecting this command you will see the following prompt on the command line:

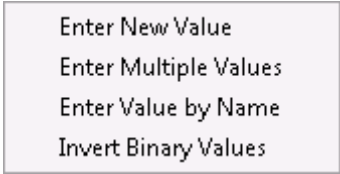
Show point:

Click the left mouse button on the desired connection points one by one. Test points are added. Stop inserting test points by pressing Return or Esc key.

Clear All Test Points

Removes all test points from the diagram.

12.2.5 Test Values - Test Value Managing Commands



Enter New Value
Enter Multiple Values
Enter Value by Name
Invert Binary Values

Test Values pull-down menu

Enter New Value

Opens a dialog in which you can enter a value to the selected test point or a dialog in which you can enter values to the selected function block configuration parameters. For more information, see chapter 8 [Entering a New Value](#).

Enter Multiple Values

Opens a dialog in which you can enter many values simultaneously to the selected test points. For more information, see chapter 9 [Entering Multiple Values Simultaneously](#).

Enter Value by Name

Opens a dialog in which you can enter a value to a test point. The target test point is selected by entering its name to the Name field of the dialog either by typing it or by selecting it from the drop-down list box. You can use this command, for example, when you want to modify the value of a test point in another control loop than the one being displayed.

Invert Binary Values

Inverts the binary state value of a binary-type test point (from 0 to 1 or from 1 to 0).

First select the menu bar command Test Values -> Invert Binary Values and then the binary-type test point to be inverted by clicking the left mouse button on it. The binary state value of the test point will be inverted.

12.2.6 Trends - Trend Managing Commands

Create Trend
Move (Pause required)
Resize Trend (Pause required)
Insert Signal as Trend Curve
Remove All Trends

Trends pull-down menu

Create Trend

Inserts a trend in the diagram. For more information, see chapter 7 [Viewing Test History as Trend Curves](#).

Move (Pause required)

Moves a trend to another location in the diagram. Note that you can use this command only when Function Test is in *Pause* state.

After selecting this command, select the trend to be moved with the left mouse button. Confirm the selection with the right mouse button. Next, use the left mouse button to specify the base point (grabbing point) used for locating the trend when moving. Then point the place where you want to paste the trend and fix it there by clicking the left mouse button or by pressing the Enter key.

If the *Ortho* mode is on, trends can be moved only in horizontal or vertical direction, not diagonally. To set the *Ortho* mode off temporarily, press the Shift key while moving the trend.

Resize Trend (Pause required)

Resizes a trend in the diagram. Note that you can use this command only when Function Test is in *Pause* state.

After selecting this command, point the mouse cursor at the trend to be resized and select it with the left mouse button. Confirm the selection with the right mouse button. Next, move the mouse cursor to specify the new size of the trend (scaling point). Finally click the left mouse button.

Insert Signal as Trend Curve

Adds a signal (test point) to the trend, after which the signal values will be shown as a trend curve. For more information, see chapter 7 [Viewing Test History as Trend Curves](#).

Remove All Trends

Removes all the trends in the diagram.

12.2.7 Test Control - Test History Managing Commands



Test Control pull-down menu

Play

Starts updating the test values in the diagram and collecting test history from the variables in the diagram. For more information, see chapter 6 [Viewing Test History as Test Point Values](#).

Pause

Pauses the updating of the test values and test history collection in the diagram. For more information, see chapter 6 [Viewing Test History as Test Point Values](#).

Set Automatic Start...

Adds the Automatic start function to the diagram. For more information, see chapter 10 [Controlling the Test with Automatic Start and Stop](#).

Set Automatic Stop...

Adds the Automatic stop function to the diagram. For more information, see chapter 10 [Controlling the Test with Automatic Start and Stop](#).

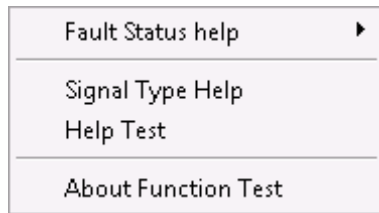
Save Test History as File...

Saves the test history as a text file in order to view all the numeric values from all the variables at one go, or to move the values to Excel a for further analysis, for example.

Clear Test History

Clears the test history and resets the collection to start from the first cycle.

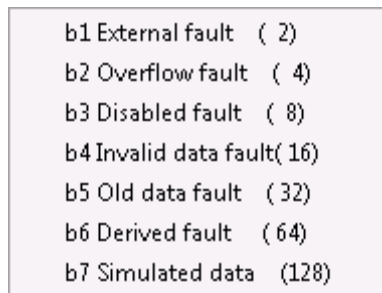
12.2.8 Help - Tool Help



Help pull-down menu

Fault Status help

The Fault Status help command opens the following submenu:



Fault Status help submenu

From the submenu, you can see an explanation (in text format) for a fault status shown in a diagram. For example, fault status 2 means *External fault*.

Signal Type Help

Gives help on types used in Function Block and Sequence CAD tools. After you have picked the command from the menu bar, the following prompt appears on the command line:

Enter type name:

Give the required type name and press Enter key. The help is shown in its own window.

Help Test

Opens the Function Test manual. This however, requires right definitions in the workspace settings in the Settings Editor tool.

About Function Test

About command brings the tool's version information on the display.

